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what is ssh?

- the secure shell protocol is cryptographic network protocol for for operating network services securely over an unsecured network. it’s most notable applications are remote login and command-line execution.
- SSH may be used in several methodologies. In the simplest manner, both ends of a communication channel use automatically generated public-private key pairs to encrypt a network connection, and then use a password to authenticate the user.
- When the public-private key pair is generated by the user manually, the authentication is essentially performed when the key pair is created, and a session may then be opened automatically without a password prompt. In this scenario, the public key is placed on all computers that must allow access to the owner of the matching private key, which the owner keeps private. While authentication is based on the private key, the key is never transferred through the network during authentication. SSH only verifies that the same person offering the public key also owns the matching private key.

Authentication: SSH keys

- on unix-like systems, the list of authorized public keys is typically stored in the home directory of the user that is allowed to log in remotely, in the file ~/.ssh/authorized_keys.
- This file is respected by SSH only if it is not writable by anything apart from the owner and root.
- when the public key is present on the remote end, and the matching private key is present on the local end, typing in the password is no longer required. however, for additional security, the private key itself can be locked with a passphrase.
- The private key can also be looked for in standard places, and its full path can be specified as a command-line setting (the option -i for ssh). The ssh-

keygen utility produces the public and private keys, always in pairs.

Generating SSH keys

Checking for existing SSH keys

- Before generating SSH keys it is best practice to check if their is any existing keys on the local machine.

1. open terminal
2. enter ls -al ~/.ssh to see if their is any ssh keys
3. If you receive an error that *~/.ssh* doesn't exist, you do not have an existing SSH key pair in the default location.
4. if their is an ssh key i can use it if you want or you could generate a new key

SSH key passphrases

- when you generate an ssh key, you can add a passphrase to further secure the key. whenever you use the key , you must enter the passphrase. if your key has a passphrase and you don’t want to enter the passphrase every time you use the key, you can add your key to the ssh agent.
- The ssh agent manages your SSH keys and remembers your passphrase.

Generating new SSH key

- open Terminal
- ssh-keygen -t ed25519 -C "email@gmail.com"
- you choose the file then you enter your passphrase and then reenter the passphrase again for confirmation

Adding SSH key to the ssh-agent

1. starting the ssh-agent in the background

```
(base) Apple@MacBook-Pro ~ % eval "$(ssh-agent -s)"
Agent pid 29833
```

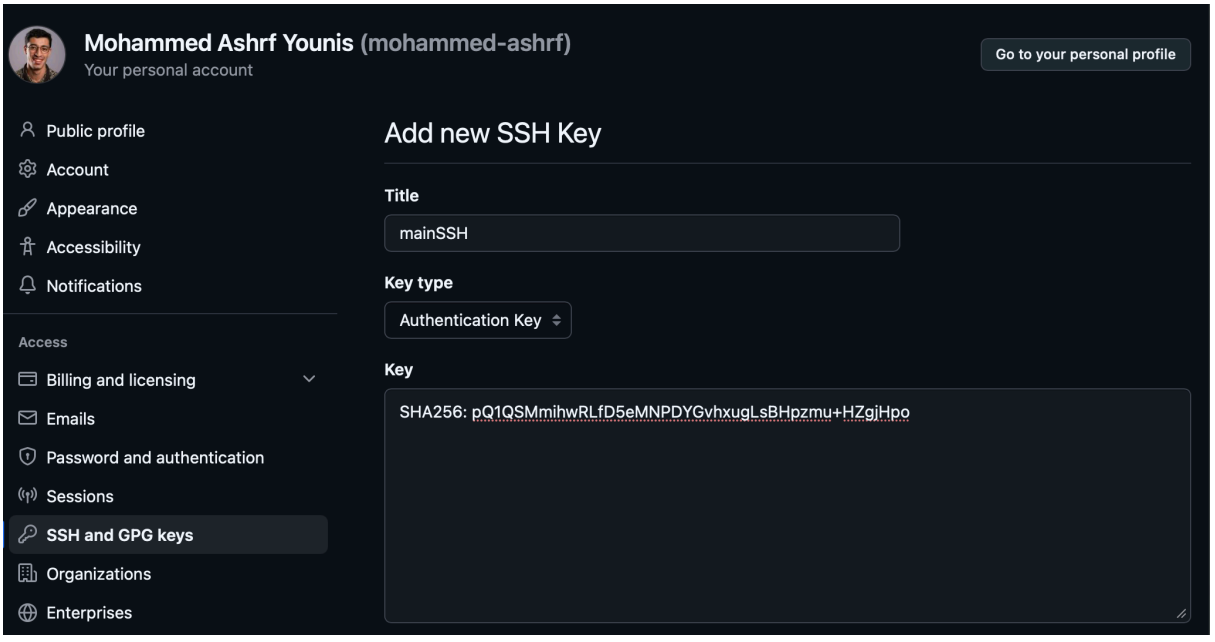
2. create the ssh config file because it doesn’t exsit

```
touch ~/.ssh/config
```

3. the ssh config should have this config

```
Host github.com
  AddKeysToAgent yes
  UseKeychain yes
  IdentityFile ~/.ssh/id_ed25519
```

4. add the ssh private key to the ssh-agent and store the passphrase in the keychain

```
ssh-add --apple-use-keychain ~/.ssh/id_ed25519
```

5. adding the public ssh key to the github account